Problem 5.13.16

Sarvesh Tamgade

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Question

Question: Let k be an integer such that the triangle with vertices

$$\mathbf{A} = \begin{pmatrix} k \\ -3k \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 5 \\ k \end{pmatrix}, \quad \mathbf{C} = \begin{pmatrix} -k \\ 2 \end{pmatrix}$$

has area 28 sq. units. Find the orthocenter of the triangle.

Solution

$$\text{Area} = \frac{1}{2}\left|\mathbf{M}\right| = 28 \tag{2.1}$$

$$\text{where} \quad \mathbf{M} = \begin{pmatrix} \mathbf{B}-\mathbf{A} & \mathbf{C}-\mathbf{A} \end{pmatrix} = \begin{pmatrix} 5-k & -2k \\ 4k & 2+3k \end{pmatrix} \tag{2.2}$$

$$\det(\mathbf{M}) = (5-k)(2+3k) - (-2k)(4k) = 10 + 13k + 5k^2 \tag{2.3}$$

$$10 + 13k + 5k^2 = \pm 56 \tag{2.4}$$

After solving,

$$k = 2 \tag{2.5}$$

Solution

$$\text{For } k = 2, \quad \mathbf{A} = \begin{pmatrix} 2 \\ -6 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 5 \\ 2 \end{pmatrix}, \quad \mathbf{C} = \begin{pmatrix} -2 \\ 2 \end{pmatrix} \tag{2.6}$$

The equation of the altitude through **A** is:

$$\mathbf{n}^T\mathbf{x} = 1 \tag{2.7}$$

This altitude passes through **A**, so

$$\mathbf{n}^T\mathbf{A} = 1 \tag{2.8}$$

$$\mathbf{n}^\top \begin{pmatrix} 2 \\ -6 \end{pmatrix} = 1 \tag{2.9}$$

and the normal vector **n** of the altitude must be parallel to $\mathbf{B} - \mathbf{C}$:

$$\mathbf{n} = t\begin{pmatrix} 7 \\ 0 \end{pmatrix} \tag{2.10}$$

Solution

Substitute to get:

$$t\begin{pmatrix}7\\0\end{pmatrix}^{\top}\begin{pmatrix}2\\-6\end{pmatrix}=1 \implies 14t=1 \implies t=\frac{1}{14} \tag{2.11}$$

Hence,

$$\mathbf{n}=\begin{pmatrix}\frac{1}{2}\\0\end{pmatrix} \tag{2.12}$$

The equation of altitude through $\mathbf{A}$ is:

$$\begin{pmatrix}\frac{1}{2} & 0\end{pmatrix}\mathbf{x}=1 \tag{2.13}$$

The equation of the altitude through $\mathbf{B}$ is:

$$\mathbf{n}^T\mathbf{x}=1 \tag{2.14}$$

This altitude passes through $\mathbf{B}$, so

$$\mathbf{n}^T\mathbf{B}=1 \tag{2.15}$$

Solution

$$\mathbf{n}^\top \begin{pmatrix} 5 \\ 2 \end{pmatrix} = 1 \tag{2.16}$$

and the normal vector $\mathbf{n}$ must be parallel to $\mathbf{A} - \mathbf{C}$:

$$\mathbf{n} = t \begin{pmatrix} 4 \\ -8 \end{pmatrix} \tag{2.17}$$

Substitute to get:

$$t \begin{pmatrix} 4 \\ -8 \end{pmatrix}^\top \begin{pmatrix} 5 \\ 2 \end{pmatrix} = 1 \implies t(20 - 16) = 1 \implies 4t = 1 \implies t = \frac{1}{4} \tag{2.18}$$

Hence,

$$\mathbf{n} = \begin{pmatrix} 1 \\ -2 \end{pmatrix} \tag{2.19}$$

Solution

The equation of altitude through $\mathbf{B}$ is:

$$\begin{pmatrix} 1 & -2 \end{pmatrix} \mathbf{x} = 1 \tag{2.20}$$

Combining equations (2.13) and (2.20):

$$\begin{pmatrix} \frac{1}{2} & 0 \\ 1 & -2 \end{pmatrix} \mathbf{x} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \tag{2.21}$$

Writing the augmented matrix:

$$\left(\begin{array}{cc|c} \frac{1}{2} & 0 & 1 \\ 1 & -2 & 1 \end{array}\right) \tag{2.22}$$

Perform the row operation $R_2 \to R_2 - 2R_1$:

$$\left(\begin{array}{cc|c} \frac{1}{2} & 0 & 1 \\ 0 & -2 & -1 \end{array}\right) \tag{2.23}$$

Solution

Therefore, the solution vector is:

$$\mathbf{x} = \begin{pmatrix} 2 \\ \frac{1}{2} \end{pmatrix} \tag{2.24}$$

Answer: The orthocentre of the triangle is

$$\boxed{\mathbf{x} = \begin{pmatrix} 2 \\ \frac{1}{2} \end{pmatrix}}$$

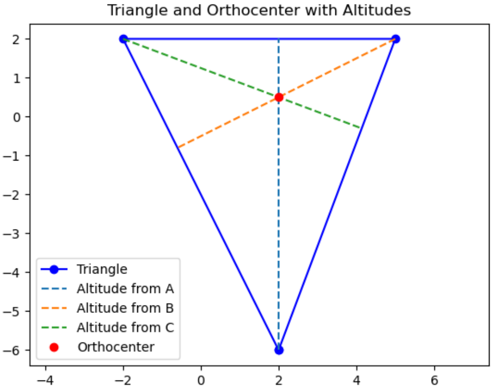

Figure: Vector Representation

C Code

```c
#include "matfun.h"

void foot_of_perpendicular(const double *P, const double *Q,
    const double *R, double *foot) {
    double QR[2] = {R[0]-Q[0], R[1]-Q[1]};
    double QP[2] = {P[0]-Q[0], P[1]-Q[1]};
    double dot_QP_QR = QP[0]*QR[0] + QP[1]*QR[1];
    double dot_QR_QR = QR[0]*QR[0] + QR[1]*QR[1];
    double t = dot_QP_QR / dot_QR_QR;
    foot[0] = Q[0] + t * QR[0];
    foot[1] = Q[1] + t * QR[1];
}

void orthocenter(const double *A, const double *B, const double *
    C, double *O) {
    double footA[2], footB[2];
    foot_of_perpendicular(A, B, C, footA);
    foot_of_perpendicular(B, A, C, footB);
```

C Code

```
    double a1 = footA[1] - A[1];
    double b1 = A[0] - footA[0];
    double c1 = a1 * A[0] + b1 * A[1];

    double a2 = footB[1] - B[1];
    double b2 = B[0] - footB[0];
    double c2 = a2 * B[0] + b2 * B[1];

    double det = a1 * b2 - a2 * b1;
    if (det == 0) { // parallel lines
        O[0] = O[1] = 0.0;
        return;
    }
    O[0] = (c1 * b2 - c2 * b1) / det;
    O[1] = (a1 * c2 - a2 * c1) / det;
}
```

```python
import numpy as np
import matplotlib.pyplot as plt
import numpy.ctypeslib as npct
from ctypes import CDLL

lib = CDLL('./libmatfun.so')

array1d = npct.ndpointer(dtype=np.float64, ndim=1, shape=(2,))

lib.foot_of_perpendicular.restype = None
lib.foot_of_perpendicular.argtypes = [array1d, array1d, array1d,
    array1d]
lib.orthocenter.restype = None
lib.orthocenter.argtypes = [array1d, array1d, array1d, array1d]

A = np.array([2.0, -6.0])
B = np.array([5.0, 2.0])
C = np.array([-2.0, 2.0])
```

Python + C Code

```
def foot_of_perpendicular(P, Q, R):
    foot = np.zeros(2, dtype=np.float64)
    lib.foot_of_perpendicular(P, Q, R, foot)
    return foot

def calc_orthocenter(A, B, C):
    O = np.zeros(2, dtype=np.float64)
    lib.orthocenter(A, B, C, O)
    return O

def plot_altitude(vertex, foot, label):
    plt.plot([vertex[0], foot[0]], [vertex[1], foot[1]],
         linestyle='--', label=label)

footA = foot_of_perpendicular(A, B, C)
footB = foot_of_perpendicular(B, A, C)
footC = foot_of_perpendicular(C, A, B)
O = calc_orthocenter(A, B, C)
```

Python + C Code

```python
# Print orthocenter
print(f"Orthocenter: {O}")

# Plot triangle
triangle_x = [A[0], B[0], C[0], A[0]]
triangle_y = [A[1], B[1], C[1], A[1]]
plt.plot(triangle_x, triangle_y, 'bo-', label='Triangle')

# Plot the altitudes
plot_altitude(A, footA, 'Altitude from A')
plot_altitude(B, footB, 'Altitude from B')
plot_altitude(C, footC, 'Altitude from C')
```

Python + C Code

```
# Plot orthocenter
plt.plot(O[0], O[1], 'ro', label='Orthocenter')

plt.axis('equal')
plt.title('Triangle and Orthocenter with Altitudes')
plt.legend()
plt.grid(True)
plt.show()
```

Python Code

```
import numpy as np
import matplotlib.pyplot as plt

A = np.array([2, -6])
B = np.array([5, 2])
C = np.array([-2, 2])

def foot_of_perpendicular(P, Q, R):
    """
    Calculate foot of perpendicular from point P to line QR.

    Returns the point (x,y) on line QR closest to P.
    """
    QR = R - Q
    QP = P - Q
    t = np.dot(QP, QR) / np.dot(QR, QR)
    foot = Q + t * QR
    return foot
```

Python Code

```python
def plot_altitude(vertex, foot, label):
    """
    Plot altitude segment from vertex to foot point.
    """
    plt.plot([vertex[0], foot[0]], [vertex[1], foot[1]],
        linestyle='--', label=label)

# Calculate feet of perpendiculars (altitudinal feet)
footA = foot_of_perpendicular(A, B, C) # from A to BC
footB = foot_of_perpendicular(B, A, C) # from B to AC
footC = foot_of_perpendicular(C, A, B) # from C to AB

# Plot triangle
triangle_x = [A[0], B[0], C[0], A[0]]
triangle_y = [A[1], B[1], C[1], A[1]]
plt.plot(triangle_x, triangle_y, 'bo-', label='Triangle')
```

Python Code

```
# Plot altitudes
plot_altitude(A, footA, 'Altitude from A')
plot_altitude(B, footB, 'Altitude from B')
plot_altitude(C, footC, 'Altitude from C')

# Compute orthocenter as intersection of two altitudes
# Use line parameter form: for line segment vertex-foot
def line_params(P, Q):
    # Line vector:
    v = Q - P
    # Line coefficients: a*x + b*y = c form
    a = v[1]
    b = -v[0]
    c = a*P[0] + b*P[1]
    return a, b, c
```

Python Code

```
a1, b1, c1 = line_params(A, footA)
a2, b2, c2 = line_params(B, footB)
A_mat = np.array([[a1, b1], [a2, b2]])
b_vec = np.array([c1, c2])

O = np.linalg.solve(A_mat, b_vec)

# Plot orthocenter
plt.plot(O[0], O[1], 'ro', label='Orthocenter')

plt.axis('equal')
plt.title('Triangle and Orthocenter with Altitudes')
plt.legend()
plt.savefig("fig1.png")
plt.grid(True)
plt.show()
```